

Smart E-Commerce Analytics Platform

Capstone Project Report

Dataset	Olist Brazilian E-Commerce + Flipkart Reviews (Kaggle)
Orders Analyzed	96,477 delivered orders (2016–2018)
Total Revenue	R\$15,422,462
Unique Customers	96,477
Product Categories	74
Date Range	2016-10-03 → 2018-08-29

Project Modules

Week	Module	Technique	Dataset
1	EDA Overview	Pandas, Matplotlib, Seaborn	Olist (9 CSVs)
2	Customer Segmentation	RFM + K-Means (k=4)	orders_clean.csv
3	Churn Prediction	Random Forest Classifier	churn_clean.csv
4	Sales Forecasting	LSTM (30-day look-back)	orders_clean.csv
5	Recommendation System	TF-IDF + Cosine Similarity	flipkart_clean.csv

1. Exploratory Data Analysis (EDA)

The Olist Brazilian E-Commerce dataset consists of 9 relational CSV files covering ~100,000 orders placed between September 2016 and October 2018. After merging and cleaning, 96,477 delivered orders were retained for analysis.

1.1 Data Cleaning Pipeline

Step	Action	Impact
1	Merged 9 Olist CSVs on order_id / customer_id / product_id	Single unified table
2	Translated Portuguese category names → English	71 categories mapped
3	Aggregated payment values per order (sum)	One row per order
4	Aggregated item counts per order	quantity column added
5	Averaged review scores per order	review_score column added
6	Filtered to order_status == delivered	99,441 → 96,477 rows
7	Dropped rows with missing total_amount or timestamp	No null revenue
8	Filled missing category → Unknown	0 nulls remaining
9	Filled missing review_score → 3.0 (neutral)	0 nulls remaining
10	Renamed columns for consistency	Standardized schema

1.2 Key EDA Findings

- Revenue grew steadily from late 2016, peaking in November 2018 (Black Friday effect)
- Top 3 categories by revenue: Health & Beauty, Watches & Gifts, Bed/Bath/Table
- São Paulo (SP) accounts for ~42% of all orders — dominant market
- Average order value: R\$154.10 | Median: R\$108.10 (right-skewed distribution)
- Credit card is the dominant payment method (~74% of transactions)
- 57% of reviews are 5-star; average review score: 4.09/5
- Peak ordering hours: 10 AM–4 PM; Monday is the busiest day

2. Customer Segmentation (RFM + K-Means)

RFM (Recency, Frequency, Monetary) analysis was performed on 96,477 orders across 96,477 unique customers. Features were log-transformed to reduce right-skew, then standardized before K-Means clustering.

2.1 Methodology

The Elbow Method and Silhouette Score analysis were used to determine the optimal number of clusters. k=4 was selected, producing four distinct customer segments ordered by average spend.

2.2 Segment Profiles

Segment	Avg Recency (days)	Avg Frequency	Avg Monetary (R\$)
Champions	244	1.0	R\$521.77
Loyal	291	1.0	R\$142.77
At Risk	42	1.0	R\$123.68
Lost	282	1.0	R\$52.29

2.3 Business Insights

- Champions: Low recency, high frequency, high spend — reward with loyalty programs
- Loyal: Regular buyers with moderate spend — upsell opportunities
- At Risk: Haven't purchased recently — target with win-back campaigns
- Lost: High recency, low frequency — consider re-engagement discounts

3. Customer Churn Prediction

A Random Forest classifier was trained on the E-Commerce Customer Churn dataset (3,941 customers) to predict churn. The dataset has a churn rate of 17.1%.

3.1 Model Performance

Metric	Value	Interpretation
Accuracy	92.9%	Overall correct predictions
F1 Score	0.7686	Harmonic mean of precision & recall
ROC-AUC	0.9579	Excellent discrimination (>0.9)
Train Size	3,152	80% stratified split
Test Size	789	20% stratified split

3.2 Top Churn Predictors

- Tenure — Customers with shorter tenure churn significantly more
- Cashback Amount — Lower cashback correlates with higher churn
- Days Since Last Order — Longer inactivity strongly predicts churn
- Complain — Customers who complained have 2x higher churn rate
- Satisfaction Score — Lower scores (1-2) show highest churn rates

4. Sales Forecasting (LSTM)

An LSTM (Long Short-Term Memory) neural network was trained on daily revenue aggregated from Olist orders. A 30-day sliding window was used to predict the next day's revenue.

4.1 Model Architecture

Layer	Units	Details
Input	30	30-day look-back window
LSTM	64	return_sequences=True
Dropout	20%	Regularization
LSTM	32	Final LSTM layer
Dropout	20%	Regularization
Dense	16	ReLU activation
Dense	1	Output: next-day revenue

4.2 Model Performance

Metric	Value
MAE (Mean Absolute Error)	R\$2,396
RMSE (Root Mean Squared Error)	R\$2,876
MAPE (Mean Absolute % Error)	7.7%

The model achieves a MAPE of 7.7%, indicating strong forecasting accuracy. Early stopping was used to prevent overfitting (patience=10 epochs).

5. Product Recommendation System

A hybrid recommendation system was built using the Flipkart Product Reviews dataset (189,869 reviews across multiple product categories).

5.1 Approaches Implemented

Approach	Method	Use Case
Popularity-Based	Weighted score: 60% review volume + 40% avg rating	Cold-start / new users
Content-Based	TF-IDF on review text + Cosine Similarity	Similar product discovery

5.2 Technical Details

- TF-IDF Vocabulary: 3,000 features, min_df=2, English stop words removed
- Products indexed: aggregated reviews per product (up to 10 reviews per product)
- Similarity matrix: cosine similarity computed across all product pairs
- Category filtering: popularity recommendations support category-level filtering

6. Interactive Dashboard & Deliverables

An interactive Streamlit dashboard was built with 5 pages corresponding to each module. The dashboard loads real data and trained models to provide live analytics.

6.1 Dashboard Pages

Page	Features
■ EDA Overview	Raw data inspection, cleaning steps, revenue trends, category/state analysis
■ Customer Segmentation	RFM computation, cluster visualization, PCA scatter, segment lookup
■ ■ Churn Prediction	Live churn prediction, feature importance, ROC curve, confusion matrix
■ Sales Forecast	30-day LSTM forecast, actual vs predicted, MAE/RMSE metrics
■ Recommendations	Popularity ranking, content-based search, similarity heatmap

6.2 Saved Model Files

File	Description
kmeans_segmentation.pkl	Trained K-Means model (k=4)
scaler_rfm.pkl	StandardScaler for RFM features
segment_summary.csv	Segment profile statistics
rf_churn_model.pkl	Random Forest classifier (200 trees)
churn_metrics.json	Accuracy, F1, AUC metrics
lstm_sales_forecast.keras	Trained LSTM model weights
scaler_timeseries.pkl	MinMaxScaler for time series
lstm_metrics.json	MAE, RMSE, MAPE metrics
tfidf_vectorizer.pkl	Fitted TF-IDF vectorizer
tfidf_matrix.npz	Sparse TF-IDF feature matrix
popularity_scores.csv	Product popularity rankings
products_index.csv	Product metadata index

6.3 Technology Stack

Category	Libraries / Tools
Data Processing	pandas, numpy
Visualization	matplotlib, seaborn, plotly
Machine Learning	scikit-learn (KMeans, RandomForest, TF-IDF)
Deep Learning	TensorFlow / Keras (LSTM)

Dashboard	Streamlit
Report	ReportLab
Data Sources	Olist (Kaggle), Flipkart Reviews (Kaggle)